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DATA VISUALIZATION TECHNIQUES

UNIT - IV

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Visualizing Cluster Analysis and Finite Mixture Models

- **Introduction**
  - The Data Sets
  - Software
- **Hierarchical Cluster Analysis**
  - Dendrograms
  - Heatmaps
- **Partitioning Cluster Analysis**
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  - The Voronoi Partition
  - Neighborhood Graphs
  - Cluster Silhouettes
  - Cluster Location and Dispersion
  - Using Background Variables
  - Self-Organizing Maps
- **Model-Based Clustering**
- **Summary**

Introduction to Visualizing Cluster Analysis

Data visualization is a critical analytical component that significantly improves our understanding of complex multivariate datasets.

- **The Synergy of Clustering and Graphics:** Cluster analysis and visualization work together to help researchers interpret computational results and identify natural groupings within data.
- **Standard Diagnostic Visualizations:** Hierarchical clustering is typically visualized using a **dendrogram**. Partitioning clustering results are often visualized through:
  - Two-dimensional projections
  - Parallel coordinate plots
- **Enhancing the Interpretation Process:** Cluster membership is commonly displayed using:
  - Distinct color schemes
  - Glyph symbols
  - Trellis displays that separate clusters into panels

Evaluating the Quality of Clustering Partitions

- **Importance of Diagnostic Tools:** Visualization techniques help verify the robustness and correctness of clustering partitions.
- **Silhouette Plots:** Silhouette plots provide a popular and mathematically rigorous way to evaluate clustering quality by measuring how well objects fit within their assigned cluster.
- **Visualizing Algorithmic Outputs:** Clustering algorithms such as **Self-Organizing Maps (SOM)** are widely used partly because their results can be easily visualized and interpreted.
- **Focus of Analytical Techniques:** This presentation explores visualization methods designed for:
  - Hierarchical clustering
  - Partitioning clustering
  - Model-based clustering

First Running Example: The “Dentitio” Dataset

- **Source Data Description:** The **Dentitio dataset** is used to demonstrate hierarchical clustering techniques.
- **Variable Characteristics:** The dataset records counts of eight types of teeth:
  - Incisors
  - Canines
  - Premolars
  - Molars

  for both the **top jaw** and **bottom jaw**.
- **Scope of Observations:** The dataset includes observations from **66 different animal species**, including bats, moles, and the opossum.
- **Analytical Purpose:** Because the variables represent simple integer counts, the dataset provides a clear and interpretable example for demonstrating hierarchical clustering.

First Running Example: The “Dentitio” Dataset

	t.inc	b.inc	t.can	b.can	t.pre	b.pre	t.mol	b.mol
Opossum	5	4	1	1	3	3	4	4
Hairy tail mole	3	3	1	1	4	4	3	3
Common mole	3	2	1	0	3	3	3	3
Star nose mole	3	3	1	1	4	4	3	3
Brown bat	2	3	1	1	3	3	3	3
Silver hair bat	2	3	1	1	2	3	3	3
Pygmy bat	2	3	1	1	2	2	3	3
House bat	2	3	1	1	1	2	3	3
Red bat	1	3	1	1	2	2	3	3
Hoary bat	1	3	1	1	2	2	3	3
…								

The first ten observations from the dentitio data: top-jaw and bottom-jaw counts of incisors, canines, premolars and molars

Second Example: 2005 German Election Data

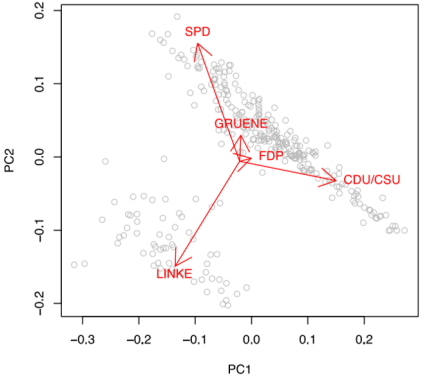
- **Context of the Electoral Data:** This dataset relates to the **German parliamentary election held on September 18, 2005**.
- **Metrics Tracked:** The dataset records the proportion of **second votes** obtained by major political parties.
- **Geographic Scope:** Voting proportions are recorded for each of the **299 electoral districts** in Germany.
- **Political Variables:** The five main parties included in the dataset are:
  - Social Democratic Party (SPD)
  - Christian Democratic Union (CDU/CSU)
  - Green Party (GRUENE)
  - Free Democratic Party (FDP)
  - The Left Party (LINKE)

Second Example: 2005 German Election Data

	SPD	CDU/CSU	GRUENE	FDP	LINKE
Flensburg–Schleswig	0.38	0.36	0.08	0.10	0.05
Nordfriesland–Dithmarschen Nord	0.36	0.41	0.06	0.10	0.04
Steinburg–Dithmarschen Süd	0.36	0.38	0.06	0.11	0.05
Rendsburg–Eckernförde	0.37	0.38	0.08	0.10	0.04
Kiel	0.41	0.28	0.13	0.09	0.06
Plön–Neumünster	0.39	0.36	0.08	0.09	0.04
Pinneberg	0.37	0.36	0.09	0.10	0.04
Segeberg–Stormarn Nord	0.36	0.36	0.09	0.11	0.04
Ostholstein	0.38	0.37	0.07	0.10	0.04
Herzogtum Lauenburg–Stormarn Süd	0.35	0.37	0.09	0.11	0.04
…					

The first ten observations from the German election data: proportions of votes for the five largest parties in each electoral district

Second Example: 2005 German Election Data



Projection of the German election data onto the first two principal components. he arrows give the projected directions of unit vectors corresponding to the five original variables

Software and Computational Environment

- **R Programming Environment:** All statistical analyses and visualizations presented here were implemented using the **R programming language**.
- **Specialized Extension Packages:** Several R packages were used to support advanced clustering and visualization capabilities.
- **Implemented Packages:** Key packages include:
  - cluster, colorspace, ellipse
  - flexclust, flexmix, fpc
  - gplots, grid, kohonen
  - lattice, mvtnorm, vcd
- **Reproducibility:** All packages are open-source and available via the **Comprehensive R Archive Network (CRAN)**. The datasets used are included in the flexclust package.

Mathematical Foundations of Hierarchical Cluster Analysis

- **The Intuitive Approach to Grouping:** Hierarchical clustering is often considered the most intuitive approach to grouping empirical data. It mirrors the logical process a human might use to divide N objects into K groups.
- **Extreme Boundary Conditions:**
  - If $K = 1$, the entire dataset forms a single cluster.
  - If $K = N$, each observation forms its own cluster.
- **Divisive Clustering Mechanics:** Divisive clustering follows a **top-down approach**. It begins with the full dataset and recursively splits it into smaller clusters until each data point forms its own group.
- **Agglomerative Clustering Mechanics:** Agglomerative clustering works **bottom-up**. It starts with N individual clusters and repeatedly merges the closest clusters until only one cluster remains.

Defining Distance Measures in Hierarchical Clustering

- **Necessity of Measurement:** Hierarchical clustering requires a mathematical distance function $d(x, y)$ to measure similarity between two d-dimensional data points.
- **Euclidean Distance:** The most widely used geometric distance:

$$d(x, y) = \sqrt{\sum_{i=1}^{d} (x_i - y_i)^2}$$

- **Manhattan Distance:** Also called the city-block distance:

$$d(x, y) = \sum_{i=1}^{d} |x_i - y_i|$$

- **Application to the Dataset:** For the **dentitio dataset**, Manhattan distance is appropriate because the variables represent integer counts of teeth.

Linkage Methods for Grouping Clusters

- **Bridging Points to Groups:** After computing distances between individual points, clustering algorithms must determine distances between entire clusters.
- **Single Linkage:**

$$l(A, B) = \min_{a \in A,\, b \in B} d(a, b)$$

- **Complete Linkage:**

$$l(A, B) = \max_{a \in A,\, b \in B} d(a, b)$$

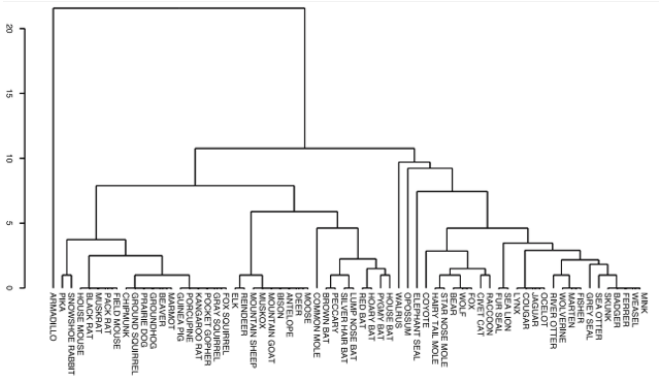
- **Average Linkage:**

$$l(A, B) = \frac{1}{|A||B|} \sum_{a \in A} \sum_{b \in B} d(a, b)$$

Understanding and Interpreting Dendrograms

- **Visualizing the Output:** Hierarchical clustering results are commonly visualized using a **dendrogram**, a tree-like structure representing cluster relationships.
- **Leaves and Branches:**
  - Leaves represent individual observations.
  - Branch heights correspond to the distance at which clusters are merged.
- **Identical Observations:** Objects with identical values appear joined at distance zero, represented by vertical lines close to their labels.
- **Layout Variability:** Branches can be flipped without changing the clustering structure. Thus, the same clustering may have up to 2^{N-1} visual layouts.

Dendrograms

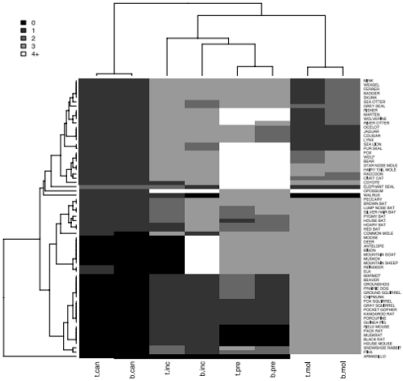


Hierarchical clustering of the dentitio data using Manhattan distance and the average linkage method

Advanced Two-Way Clustering with Heatmaps

- **Clustering Multiple Dimensions:** Hierarchical clustering can be applied not only to observations but also to variables.
- **Heatmap Visualization:** A heatmap displays the dataset as a colored matrix where each cell represents a specific value.
- **Structural Permutations:** Rows and columns are reordered based on hierarchical clustering of observations and variables.
- **Revealing Data Patterns:** In the dentitio dataset, heatmaps reveal patterns such as:
  - Similarity between top and bottom jaw tooth counts
  - Blocks representing predator species with larger incisors

Advanced Two-Way Clustering with Heatmaps



Heatmap of the dentitio data using Manhattan distance and the average linkage method for both columns and rows

Introduction to Partitioning Cluster Analysis

- **Principle of Complexity Reduction:** Partitioning cluster analysis reduces complexity by transforming multidimensional data into a single categorical variable representing **cluster membership**.
- **Analytical Objectives:** The objective is either:
  - To discover naturally occurring homogeneous subgroups in the data.
  - To impose useful mathematical groupings for analytical purposes.
- **Divide and Conquer Strategy:** Researchers apply partitioning methods to divide large datasets into smaller segments that are easier to interpret and analyze.
- **Common Algorithms:**
  - K-Means Clustering
  - Partitioning Around Medoids (PAM)

Mathematical Objectives of Partitioning Algorithms

- **Defining the Centroids:** Given a dataset

$$X_N = \{x_1, x_2, \dots, x_N\}$$

the algorithm defines a set of cluster centroids:

$$C_K = \{c_1, c_2, \dots, c_K\}$$

- **Identifying the Closest Centroid:**

$$c(x) = \arg\min_{c \in C_K} d(x, c)$$

- **Cluster Definition:**

$$A_k = \{x_n \mid c(x_n) = c_k\}$$

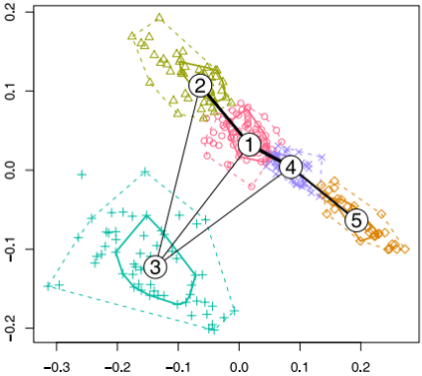
- **Optimization Objective:**

$$D(X_N, C_K) = \frac{1}{N} \sum_{n=1}^{N} d(x_n, c(x_n)) \to \min_{C_K}$$

Defining Boundaries with Convex Cluster Hulls

- **Limitations of Spanning Ellipses:** Elliptical boundaries may produce misleading cluster representations, especially for non-Gaussian data or non-Euclidean metrics.
- **Arbitrary Convex Shapes:** Empirical clusters often form irregular convex shapes that cannot be properly represented using simple parametric boundaries.
- **Nonparametric Alternative:** Convex cluster hulls provide a flexible visualization technique based on actual distances from cluster centroids.
- **Boxplot Analogy:** The visualization concept is similar to a boxplot:
  - Inner region represents the cluster core
  - Outer region represents the broader cluster spread

Convex Cluster Hulls



Convex hulls of the five clusters

Mathematics of Convex Cluster Hulls

- **Median Distance:**

$$m_k = \text{median}\{d(x_n, c_k) \mid x_n \in A_k\}$$

- **Inner Cluster Area:** The cluster core contains all points satisfying:

$$d(x_n, c_k) \leq m_k$$

- **Outer Cluster Boundary:** The extended cluster region includes points satisfying:

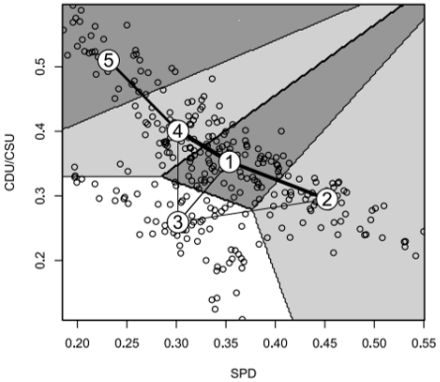
$$d(x_n, c_k) \leq 2.5\, m_k$$

- **Outlier Detection:** Points lying outside the $2.5\, m_k$ boundary are considered **structural outliers**.

Voronoi Partition and Input Space Projection

- **Partitioning the Entire Space:** The **Voronoi partition** divides the entire input space based on proximity to cluster centroids.
- **Assignment Rule:** Each point in the space is assigned to the nearest centroid according to the distance function.
- **Visualization Challenge:** Because the partition exists in high-dimensional space, visualization usually involves projecting the data into two dimensions.
- **Practical Example:** In the German election dataset, Voronoi partitions can be visualized using a 2D projection defined by voting proportions for the SPD and CDU/CSU parties.

The Voronoi Partition



Projection of the clusters into the plane spanned by SPD and CDU/CSU, and the corresponding Voronoi partition

Using Neighborhood Graphs for Structural Borders

- **Understanding Structural Relationships:** Neighborhood graphs help visualize how clusters interact along their boundaries.
- **Nodes and Edges:** Each cluster centroid forms a node. Two nodes are connected by an edge if there exists a data point for which those centroids are the closest and second-closest centers.
- **Revealing Border Interactions:** These connections reveal structural borders and the degree of interaction between clusters or segments.
- **Highlighting Projection Artefacts:** If clusters appear to overlap in a 2D projection but the neighborhood graph shows no connecting edge, the overlap is merely a projection artefact.

Calculating and Utilizing Shadow Values

- **Second-Closest Centroid:**

$$\tilde{c}(x) = \arg \min_{c \in C_K \setminus \{c(x)\}} d(x, c)$$

- **Shadow Value Formula:**

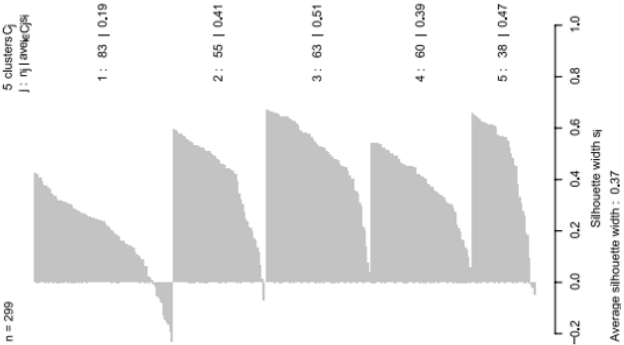
$$s(x) = \frac{d(x, c(x)) + d(x, \tilde{c}(x))}{2d(x, c(x))}$$

- **Interpretation:**
  - $s(x) \approx 0 \rightarrow$ point lies close to its cluster centroid
  - $s(x) \approx 1 \rightarrow$ point lies near the boundary between clusters
- **Cluster Boundary Strength:** Average shadow values s_{ij} can be computed for all boundary points shared by clusters i and j.

Cluster Silhouettes for Diagnostic Validation

- **High-Dimensional Challenge:** In high-dimensional datasets, it becomes difficult to visually confirm whether clusters are truly separated.
- **Diagnostic Tool:** Silhouette plots provide a statistically rigorous method for evaluating clustering quality.
- **Evaluation Mechanism:** Each point's average distance to points in its own cluster is compared with distances to points in the nearest neighboring cluster.
- **Interpretation:**
  - Large positive values → well-clustered points
  - Values near zero → border points
  - Negative values → possible misclassification

Cluster Silhouettes



Silhouette plot for the five-cluster PAM partition of the German election data

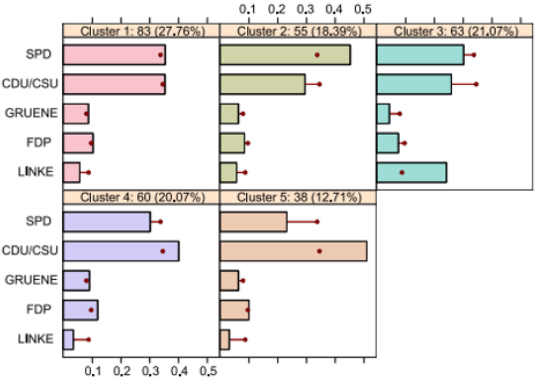
Silhouette Value Formula and Interpretation

- **Silhouette Value Formula:**

$$\mathrm{sil}(x) = \frac{b(x) - a(x)}{\max(a(x), b(x))}$$

- **Within-Cluster Similarity:** $a(x)$ represents the average dissimilarity between point x and all other points within the same cluster.
- **Neighboring Cluster Similarity:** $b(x)$ represents the smallest average dissimilarity between x and points belonging to the nearest neighboring cluster.
- **Cluster Quality Assessment:** Clusters with predominantly large positive silhouette values indicate strong separation and reliable clustering results.

Cluster Location and Dispersion



Barplot of the five-cluster medoids in comparison to the overall population mean. he numbers in the title strips of the panels give the absolute and relative number of points in each cluster

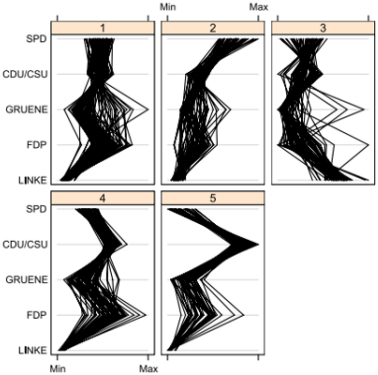
Analyzing Cluster Location and Dispersion

- **Beyond Simple Projections:** Silhouette plots often reveal stronger separation in high-dimensional space than simple projections indicate.
- **Centroid Barplot Method:** Instead of visualizing individual points, cluster centroids are displayed as bars to summarize cluster characteristics.
- **Population Comparison:** Cluster centroids are plotted alongside the overall population mean to highlight differences across variables.
- **Magnitude Interpretation:** For example, in election data, taller bars indicate significantly larger vote shares for specific parties within particular clusters.

Parallel Coordinate Plots and Trellis Displays

- **Visualizing High-Dimensional Flow:** Parallel coordinate plots allow simultaneous visualization of all variables in a high-dimensional dataset.
- **Preventing Visual Masking:** To avoid clutter caused by overlapping lines, these plots are often displayed using a **trellis display**, where each cluster is shown in a separate panel.
- **Revealing Structural Correlations:** Parallel coordinate plots can reveal hidden relationships between variables. For example, a strong negative correlation appears as a distinctive **crossing (X-shaped)** pattern between two axes.
- **Asymmetric Projections:** Researchers may also construct asymmetric projections designed to maximize the visual separation of one cluster from others.

Parallel Coordinate Plots



Parallel coordinate plot of the five clusters

Integrating Background Variables into Cluster Visualization

- **Leveraging External Context:** Cluster analysis often involves examining whether additional contextual variables show different distributions across clusters.
- **Background Variables:** These variables exist in the dataset but were intentionally excluded from the clustering process.
- **Clusters as Nominal Variables:** Cluster membership can be treated as a nominal variable and cross-tabulated with background variables.
- **Visualization Methods:**
  - Balloon plots
  - Mosaic plots

  These visualizations reveal relationships between clusters and external variables.

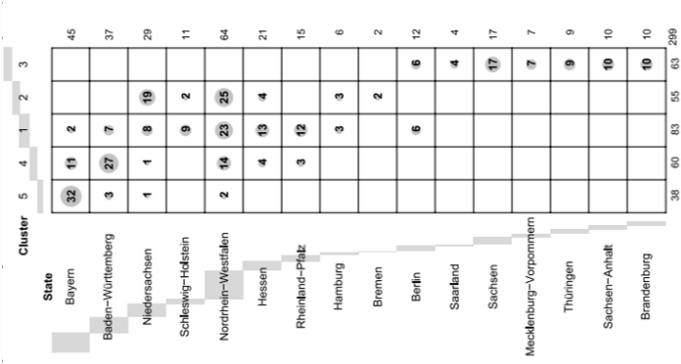
Background Variables

The 16 federal states of Germany

Structural Constancy via Self-Organizing Maps (SOMs)

- **Predetermined Grid Structure:** Self-Organizing Maps impose a predefined grid (often 2D) on cluster centroids before clustering begins.
- **Neighborhood Preservation:** The algorithm forces neighboring centroids on the grid to remain close throughout the clustering process.
- **Visualization Advantage:** Because the grid is inherently two-dimensional, results can be visualized clearly without overlapping projection artefacts.
- **Pie Chart Representation:** Cluster centroids in SOM grids are often represented using pie charts where each slice indicates the relative strength of variables.

Self-Organizing Maps (SOMs)



Cross-tabulation of German federal states and electoral district clusters, sorted by decreasing CDU/CSU vote share. Gray circle size represents table values; top and left bar heights show column and row totals.

Mathematical Foundations of Model-Based Clustering

- **Probabilistic Framework:** Model-based clustering fits a set of K probabilistic models to the dataset.
- **Finite Mixture Model:**

$$h(y|x, w) = \sum_{k=1}^{K} \pi_k f(y|x, \theta_k)$$

- **Prior Probability Constraint:**

$$\pi_k \geq 0, \quad \sum_{k=1}^{K} \pi_k = 1$$

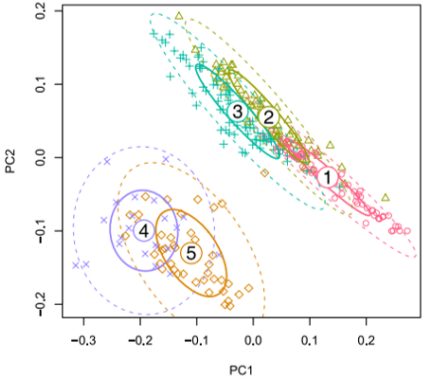
- **Gaussian Mixture Models:** In many applications, f represents a multivariate Gaussian distribution where parameters include:

$$\theta_k = (\mu_k, \Sigma_k)$$

Visualizing Model-Based Clustering and Summary

- **Confidence Ellipses:** Gaussian mixture components are visualized using elliptical confidence regions:
  - Solid ellipse → 50% confidence region
  - Dashed ellipse → 95% confidence region
- **Posterior Probability Analysis:** Rootograms are used to visualize posterior probabilities of cluster membership.
- **Summary of Hierarchical Methods:** The **dendrogram** remains the most intuitive visualization for hierarchical clustering.
- **Summary of Partitioning Methods:** Partitioning and model-based clustering rely on diagnostic plots such as:
  - Silhouette plots
  - Rootograms
  - Trellis displays

Model-Based Clustering



A five-cluster finite mixture model for the German election data projected on the first two principal components